Michael Remington Coulter

Buda, TX | mrcoulter45@gmail.com | (760) 694-6310
linkedin.com/in/michael-coulter-164868153 | github.com/mrcoulter45 | mrcoulter45.github.io/portfolio

PROFESSIONAL EXPERIENCE

Arm | Austin, TX

Senior Software Engineer Apr 2023 - Present

Interconnect Architect | CMN GUI Lead | Tools for internal engineers and external customers.

Recognition: Most Innovative Award, Engineering AI Showcase (Dec 2025), for a project-agnostic multi-agent orchestration and context-gathering system.

- Designed a modular AI engineering platform with four project packs, separating shared tooling from product-owned runtimes through manifest-driven configuration.
- Implemented Python Model Context Protocol (MCP) planning and rollback for two bounded interconnect workflows; recorded validation lists 5,063 full-suite passes and 92 live checks.
- Built isolated workspaces with dedicated ports, caches and editor state; coordinated concurrent checkout allocation through atomic locks, tokenized leases and ownership-checked recovery.
- Built Jenkins/LSF repair tooling for exact-revision replay and binary patch transfer; required three consecutive clean replays of the latest patch before finalizing flaky-test fixes.
- Built resumable RHEL 8/LSF release automation with archive checks and packaged acceptance, preserving payload hashes/permissions and verifying source/dependency provenance.
- Built immutable macOS/Windows modules and isolated AI-agent verification in fresh sessions, checking payload integrity, MCP registration, provider routing and runtime cleanup.

Software Engineer Apr 2020 - Apr 2023

- Implemented Flask services to start and stop Jupyter notebooks and persist files in AWS EFS for Pelion's embedded-ML platform; defined PostgreSQL schemas and built end-to-end tests.
- Built sensor-ingestion and dashboard APIs with InfluxDB, including a fixed-radius neighbor algorithm for workspace proximity detection.

Graduate Software Engineer Jan 2019 - Apr 2020

- Built Verilog FPGA modules and Mbed OS firmware for I2C, GPIO, ADC, DAC and PWM; contributed to shield/form-factor PCBs. 2020 Arm Quality Award (team).
- Built Node.js/Express REST APIs, PostgreSQL schemas and React dashboards to monitor occupancy from thermal cameras and PIR sensors.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Rust, SQL
AI / developer systems: MCP, typed plans, rollback, session isolation, Git, Jenkins, Playwright
Platforms: LSF, RHEL 8, macOS, Windows; React, Tauri 2, Flask, PostgreSQL

EDUCATION

The University of Texas at Austin
Bachelor of Science, Electrical and Computer Engineering | Dec 2018
Primary focus: Computer Architecture and Embedded Systems | Secondary: Integrated Circuits
1st Place (Industry), ECE Capstone Design Award - IoT Wireless Home Security System (2018)